Capstone Milestone Report: Data Cleaning, Feature Engineering, and Exploratory Data Analysis

Overview

Since the last milestone I expanded the scope of my dataset by incorporating box office revenue into the project. Although the current box office dataset is incomplete and only contains the top 1,000 highest grossing movies, I felt it was worth including because it allowed me to improve my target variable. My research question is trying to determine what characteristics make a book a good candidate for a movie adaptation. I realized there is no single objective definition of a "good" adaptation, so instead of relying on one metric I created a weighted target variable that combines critical reception, audience reception, and commercial success. I believe this creates a more robust target for regression than using any individual metric by itself.

To combine these different measurements, I converted each component into percentile rankings instead of using the raw values. I made this choice for two reasons. First, the variables exist on completely different scales. Review scores range from 1-5 while box office revenue differs by orders of magnitude. Converting everything to percentiles normalizes the variables onto the same scale and makes weighting each component much easier. Second, it makes the final target much more interpretable. A score close to 1 represents an excellent commercial adaptation opportunity while a score close to 0 represents a poor one, making it much easier to compare books and draw conclusions.

Handling Missing Values

Most of the missing values were removed naturally as a result of the features I decided to exclude. A large number of rows from the Goodreads dataset were removed because I wanted to focus on textual information instead of popularity-based variables. One of my biggest concerns was creating a chicken-and-egg problem. Variables such as author popularity, book ratings, or number of reviews may have been influenced by a successful movie adaptation rather than representing characteristics of the book itself. Since removing those variables also removed most of the missing values, I did not have to make many additional imputation decisions.

The primary exception was the box office data. Since my dataset only contains the top 1,000 highest grossing movies, many adapted books did not have a matching box office value. Instead of removing these observations, I filled the missing values with zero. This was done to prevent unnecessary data loss while still preserving the ordinal information available in the dataset. I know that movies missing from the top 1,000 made less money than those included, but I do not know their exact ordering. By assigning a value of zero before converting to percentiles, all movies outside the top 1,000 naturally fall into the lowest commercial success group without artificially ranking them against one another. This does introduce a limitation because the dataset is not adjusted for inflation, which likely disadvantages older movies, but I felt this approach was preferable to removing a large portion of the available data.

Feature Engineering

The majority of my feature engineering focused on representing the story itself rather than popularity. I intentionally removed variables such as TMDB vote averages, vote counts, Goodreads ratings, and review counts because those measures were already incorporated into my target variable. Including them as predictors would introduce unnecessary noise and could potentially leak information from the target back into the model.

The primary text features were generated using TF-IDF followed by Principal Component Analysis (PCA). My main goal was to reduce noise and reduce the overall number of features while still preserving the important information contained within the movie overviews. I specifically chose PCA because it also allows me to examine the resulting components afterward and determine which words contribute positively or negatively to each component. I did not want my conclusions to end with a prediction from a complete black box. Even if dimensionality reduction costs a small amount of predictive power, I think the additional interpretability is valuable for answering my research question.

Genre information was another feature I believed would be useful. In many ways, genres act as a human-made classifier that has already performed some semantic grouping of stories. Including genre allows me to capture broad thematic information without relying entirely on the raw text.

I also included description length as a feature under the hypothesis that longer descriptions may represent either more complex stories or stories that publishers felt were worth describing in greater detail. However, after reviewing the engineered features I am less confident that this variable provides meaningful predictive signal and may ultimately remove it if it proves to be mostly noise.

Modeling Approach

Although this milestone focuses primarily on exploratory data analysis and feature engineering, I chose to train a baseline Random Forest model to complete the entire machine learning pipeline. My goal was not to optimize model performance at this stage but rather to verify that the preprocessing, feature engineering, and target construction all worked together correctly. This establishes a baseline that I can improve upon during later milestones when model refinement becomes the primary focus.

I also chose to formulate the problem as a regression task instead of a classification problem. Creating a continuous adaptation score allows the finished model to rank books that have never been adapted into movies, something that would not be possible with a binary classifier. I also think it provides greater flexibility for decision-making. Different producers have different budgets and different levels of acceptable risk, so a continuous score provides more information than simply labeling a book as a good or bad adaptation candidate.

Exploratory Data Analysis

The exploratory analysis focused on understanding the distributions of the engineered variables, identifying potential outliers, and examining relationships between features before moving into modeling. Summary statistics, distribution plots, boxplots, correlation heatmaps, and feature importance visualizations were generated to better understand the data.

Rather than simply generating plots, the goal of this analysis was to evaluate whether the engineered features captured meaningful information and to identify features that may need to be revised or removed before future modeling. This process also helped validate many of the design decisions made during preprocessing and feature engineering.

Next Steps

I am still working on cleanly integrating the wikipedia articles as they are giving me some trouble with accuracy. I also want to fold in more box office data and adjust for inflation but I may have to do that myself. The pipeline I have of feature engineering to model testing should allow for me to fold in more boxoffice or movie title information in the future with minimal difficulty.